

Cheat Sheet: Getting Started with MCP

Estimated time needed: 10 minutes

This module explores the basics of creating and configuring MCP servers. Use this cheat sheet to quickly review and refresh the concepts you learned.

MCP basics

Topic	Description
What is MCP?	Model Context Protocol (MCP) is an open-source standard for enabling AI applications with the capabilities of external systems. Using MCP, large language models (LLMs) like ChatGPT can execute tasks (e.g., searching the web, retrieving data (e.g., calendar data from the local file system), and perform specialized workflows (e.g., workflow rule engine).
Analogy	A simple analogy for MCP is to think of it like a USB port for AI applications. MCP provides a standardized interface that lets AI applications connect with external systems, just like how USB offers a convenient way for devices to connect and communicate.
Why MCP?	Seamless integration: Red a standardized way for connecting AI to external systems, consistent and simple development patterns improve understandability and scalability. Interoperability: Wide support for different clients and AI models, and a modular Security: Uses OAuth 2.0 and token-based authentication. Efficiency: Connects to external systems that are kept up-to-date, ensuring clean, accurate data, and robust, consistent responses. Agentic workflow support: Enables AI agents for multi-step, dynamic tasks.

MCP architecture

Topic	Description
Three-layer architecture	MCP uses bidirectional communication using three message types in JSON-RPC 2.0 (Communication format). These messages are sent over STDIO or HTTP , transports you will learn in the next module.
Initialization	1. InitializationClient sends a initialize request with protocol version and capabilities. Server responds with its capabilities. Client sends a capabilities notification to complete the handshake. 2. ToolInvocation : Client sends a callTool request to invoke a tool. Server responds with the result. The response is sent back to the client. 3. Resource : Client sends a getResources request to retrieve resources. Server responds with the list of resources. The response is sent back to the client.